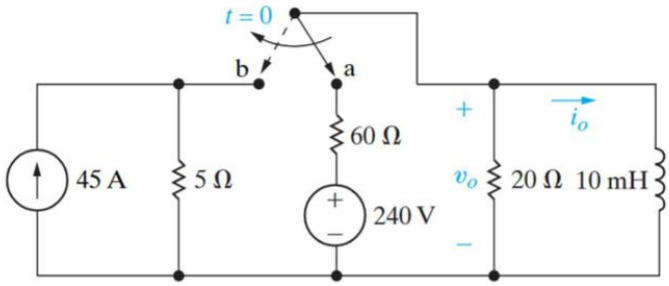
EE 2010
Fall 2026

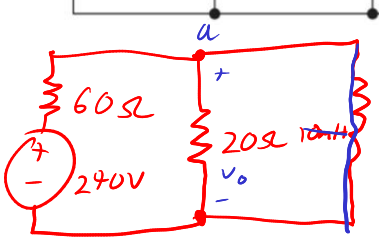
Name _____

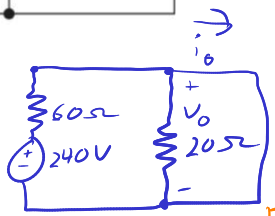
Game 39
10/26/2026

Find

a) $i_o(t = 0^-)$
b) $i_o(t = 0^+)$
c) $i_o(t = \infty)$
d) $i_o(t)$

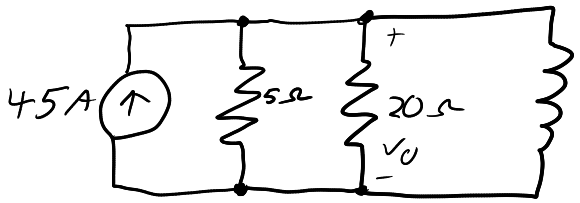






$v_o = 0V$
$i_{20} = 0A$

$i_{tot} = \frac{240V}{60\Omega} = 4A = i_o(0_-)$



$= 4A$

$i_o(t=\infty) = 45A$

$\frac{v_o}{5} + \frac{v_o}{20} + 4 = 45$

$\frac{v_o}{4} = 41 \quad v_o = 164V$

$i_o = i_L + 8.2 = \boxed{12.2A}$

$\tau = \frac{0.01}{4}$

$5\Omega \,||\, 20\Omega \quad R_{eq} = 4\Omega \quad \frac{1}{\tau} = 400$

$i_o(t) = i_o(\infty) + [i_o(0_+) - i_o(\infty)]\, e^{\frac{-t}{\tau}}$

$= \boxed{45 - 32.8e^{-400t}A}$